

DATA PREPARATION

ADDED FEATURES

	TARGET_BY_YEAR
	TARGET_BY_TERM
	DATA_BY_TERM
	SINGLE_POINT_DATA
	ALL_DATA_6162023.parquet

All data begins with the raw data created in the Data wrangling folder. The data is called the “ALL_DATA_6162023”. This Python script encompasses the entire process of adding additional features and generating various versions of the target variables. It includes sections where the data is segmented by student level to facilitate proper aggregation and labeling of the features.

DISTRIBUTION OF STUDENT LEVEL

```
df["STUDENT_LEVEL_DESC"].value_counts()
```

```
Undergrad - Chicago      423497
Graduate - Chicago      176893
Professional - Chicago   83388
Graduate Non-Degree Chicago 14320
Undergrad Non-Degree Chicago 9717
Law - Chicago            6995
Non-Credit - Chicago    1650
Graduate Online - Chicago 767
SCALES - Chicago        111
Name: STUDENT_LEVEL_DESC, dtype: int64
```

For example, when creating a column that represents the number of terms completed, a decision was made to ensure its relativity to the student's current degree pursuit. This approach was adopted to avoid situations where a student might have both undergraduate and graduate terms, leading to inflated graduate term counts for some individuals. By maintaining the terms'

relativity to the current degree, the model can effectively learn the correlation between the number of completed terms and the likelihood of degree completion within that specific degree level.

UNIQUE_TERM_CD	AH_ACAD_YEAR_CD_LIST	TOTAL_COMPLETED_TERMS	CURRENT_COMPLETED_TERMS	CURRENT_COMPLETED_ACAD_YEARS
220218,220221,220228,220235	2122,2223	4	1	1
220218,220221,220228,220235	2122,2223	4	2	1
220218,220221,220228,220235	2122,2223	4	3	2
220218,220221,220228,220235	2122,2223	4	4	2
220135,220138	1213,1314	2	1	1

UNIQUE_TERM_CD

This information provides a comprehensive record of each term in which the student was enrolled. Later on, it serves the purpose of labeling a student as "continuing." This is achieved by examining the current term in a given data point and cross-referencing it with this list to determine if the student was enrolled in the subsequent term. If the student was enrolled, it is labeled as 1; otherwise, it is labeled as 0. Additionally, this list of terms is utilized to calculate the count of completed terms.

AH_ACAD_YEAR_CD_LIST

This dataset includes the academic year codes for each academic year in which the student was enrolled. The utilization of year codes is preferred over simple calendar years due to the academic context. Typically, a full year of schooling is considered to extend from the fall of one year to the summer of the following year. Therefore, using academic year codes provides a more suitable representation for tracking and analyzing academic progress.

TOTAL_COMPLETED_TERMS

The term "Total completed terms" encompasses all terms completed throughout the recorded history. While it can provide useful information, including this column in training and testing machine learning models may result in overfitting. It is recommended to exclude this column from the model training process, as its inclusion might potentially reveal the answer and hinder the model's ability to generalize. However, this column was utilized for statistical analysis purposes to gain insights into the data but is not recommended for training the model.

CURRENT_COMPLETED_TERMS

The "Current completed terms" column denotes the number of terms a student has successfully completed up to the present moment. It is highly recommended to retain this column for training

purposes as it can be instrumental in identifying general trends regarding students' likelihood of discontinuation. It appears that early on in a student's academic journey, there is a higher risk of discontinuation, whereas as they progress closer to completing their coursework, the likelihood of finishing increases. This column can provide valuable insights into the factors influencing student persistence and help in developing effective strategies for supporting their academic progress.

CURRENT_COMPLETED_ACAD_YEARS

Similar as the explanation above, this column contains the number of completed academic years.

DATA TABLE EXPORT: DATA BY TERM

After the additional features have been appended the first export is generated. This is the data with the added features, without having any target variable added yet.

LABELING THE TARGET VARIABLE: An Explanation and Discussion

TARGET_ENROLLED_NEXT_TERM	SUCCESSFUL_OUTCOME_TARGET_BY_TERM	TARGET_ENROLLED_NEXT_YEAR	SUCCESSFUL_OUTCOME_TARGET_BY_YEAR
1	1	1	1
1	1	1	1
1	1	1	1
0	0	0	0
1	1	1	1

Considerable thought was given to determining the most suitable approach for labeling the target variable, and I would like to share my personal findings and key insights on this matter. There are three potential labels for student outcomes: discontinuation, graduation, and continuation. When it comes to discontinuation, the point at which a student does not return to classes requires careful consideration. Should we only consider the next term? If we include the next term, should we also include the summer? Not enrolling in the summer may have minimal impact on the likelihood of a student enrolling in the fall. Therefore, it is worth considering whether a student enrolled in the next term, the next fall term, or at any point in the following years. Various variations of these considerations were created, tested, and taken into account. Additionally, when examining the other two outcomes of graduating and continuing, there are both similarities and differences. Continuing implies that the student will return in the next

academic year, while graduating indicates that they will not. These outcomes involve one group leaving and the other staying. However, both outcomes share the similarity of being successful students. Consequently, different approaches were considered for labeling the target variable.

One variation involves labeling continuation as 1 and graduation as 2. Another variation labels both outcomes as 1, treating the target variable as a binary indicator of success (1) and failure, dropout, or discontinuation as (0).

The key question revolves around which of these outcome distinctions would exhibit the most similarities in terms of feature sets. It is suggested that differentiating between successful and unsuccessful outcomes would yield better results than distinguishing between continuation and discontinuation alone. Although both dropout and graduation impact total enrollment negatively, they do not share similar characteristics and including them as separate categories could weaken the predictive power of their respective feature sets.

See below the different distributions of different labeling methods. Please see the python notebook for more information on specifics of the labeling process.

```
graduate_df["TARGET_ENROLLED_NEXT_TERM"].value_counts()
```

```
1    150396
2     31047
0     10537
Name: TARGET_ENROLLED_NEXT_TERM, dtype: int64
```

```
graduate_df["TARGET_ENROLLED_NEXT_YEAR"].value_counts()
```

```
1    135640
2    44897
0     11443
Name: TARGET_ENROLLED_NEXT_YEAR, dtype: int64
```

```
graduate_df["SUCCESSFUL_OUTCOME_TARGET_BY_YEAR"].value_counts()
```

```
1    180537
0     11443
Name: SUCCESSFUL_OUTCOME_TARGET_BY_YEAR, dtype: int64
```

```
graduate_df["TARGET_ENROLLED_NEXT_FALL_YEAR"].value_counts()
```

```
1    132146
2     47019
0     12815
Name: TARGET_ENROLLED_NEXT_FALL_YEAR, dtype: int64
```

LABEL TARGET VARIABLES FOR STUDENTS THAT ENROLLED IN THE NEXT TERM

Continuation is labeled as 1 if the student enrolled in the next spring term after the fall term, and if they enrolled in the next fall term after the spring term. Additionally, if the student enrolled in the next fall term after the summer term, it is also considered as continuation. However, we do

not label the spring term as enrolled in the summer term since it does not provide meaningful information for our purposes.

- 0: Dropped Out
- 1: Continued to Next Fall Term
- 2: Graduated

LABEL TARGET VARIABLES FOR STUDENT SUCCESS

This process generates a target variable that enables the classification of students as successful or not. A value of 0 indicates that they discontinued their studies at some point, while a value of 1 signifies that they either graduated or were continuing their education.

Based on the Above Data Table

- 0: Not Successful
- 1: Successful (Continued or Graduated)

LABEL TARGET VARIABLES FOR STUDENTS THAT ENROLLED IN THE NEXT YEAR ANY TERM THAT YEAR

This step examines the current term and checks if the student has enrolled in any term within the following year. For instance, if a student enrolled in the fall of 2018 and did not enroll again until the spring of 2020, they would still be considered as continuing because they enrolled in the next year.

- 0: Dropped Out
- 1: Continued to Any Term Next Year
- 2: Graduated

LABEL TARGET VARIABLES FOR STUDENT SUCCESS BY NEXT YEAR

This creates a target variable that allows the classification for students that were successful or not. This 0 means they discontinued at some point. The 1 means they either graduated or were continuing.

Based on the Above Data Table

- 0: Not Successful
- 1: Successful(Continue or Graduated)

LABEL TARGET VARIABLES FOR STUDENTS THAT ENROLLED IN THE NEXT YEAR BUT ONLY FALL SEMESTER

This step examines the current term and checks if the student has enrolled in the next fall term in the following year. For example, if a student enrolled in the fall of 2018, the process would look for the fall term in 2019. If it exists, the student would be labeled as 1. Similarly, if the current term code is in the spring of 2019, the process would look for the fall term code in 2019. If it exists, the student would also be labeled as 1.

- 0: Dropped Out
- 1: Given Any Given Term of Current Year, They Continued to Only Next Fall Term
- 2: Graduated

AGGREGATION METHODS

SINGLE POINT PER TERM: TARGET BY TERM

This is the data with the labeled target variables that has not undergone any aggregation methods. Can be used for prediction for term over term predictions.

SINGLE POINT PER YEAR: TARGET BY YEAR

This data has been aggregated taking only the most recent snapshot in a year for a single student. This can be used to make predictions year over year. Grouping the data by EDW_PERS_ID and ACAD_YEAR, we aggregate the information by selecting the most recent semester for each academic year. This process aims to reduce the number of snapshots for each student. For example, if a student was enrolled in the fall and spring of 2013, only the snapshot from the spring term would be retained. Similarly, if a student was enrolled in the fall, spring, and summer, only the snapshot from the summer term would be kept. In cases where a student was only enrolled in the fall, only the fall term snapshot would be retained.

SINGLE POINT DATA: DOWNSAMPLED RANDOM SELECTION

To prevent overfitting of models, it became evident that efforts should be made to avoid having data points that are nearly duplicates. When observing students over time, certain features remain unchanged throughout their academic journey. For example, attributes such as sex, degree, resident status, first-generation status, and others would remain constant across multiple terms. Utilizing such data points would result in the majority of their features remaining unchanged, potentially giving away the answer to the training model.

To address this concern, a dataset was created to include each student only once, ensuring equal representation of individuals and their feature sets across the entire sample population. To achieve this, a random term was selected for each student. Initially, other approaches such as using the last term or maximum term for each student were considered, but these methods disproportionately included only those students who had either graduated or dropped out, as they were the last terms in which they were enrolled. By randomly selecting a term for each

student, the resulting sampling represents the same distribution of the target variable as the actual population.

This dataset was specifically created with the primary purpose of training the model. It is highly recommended to use this data table for training machine learning models, as it addresses the issue of duplicates and provides a representative sample for effective model training.